where $d_i = \dim V_i$. By Schur's lemma,

$$\mathrm{Hom}_{k[G]}(k, k[G]) = k\Lambda,$$
$$\mathrm{Hom}_{k[G]}(k[G], k) = k\epsilon,$$

for nonzero homomorphisms of representations $\epsilon : k[G] \to k$ and $\Lambda : k \to k[G]$ unique up to scaling. We can take ϵ such that $\epsilon(g) = 1$ for all $g \in G$, and we can take Λ such that $\Lambda(1) = \sum_{g\in G} g$. Then

$$\epsilon \circ \Lambda(1) = \epsilon\left(\sum_{g\in G} g\right) = \sum_{g\in G} 1 = |G|.$$

If $|G| = 0$, then Λ has no left inverse, as $(a\epsilon) \circ \Lambda(1) = 0$ for any $a \in k$. This is a contradiction. □

Example 4.1.3. If $G = \mathbb{Z}/p\mathbb{Z}$ and k has characteristic p, then every irreducible representation of G over k is trivial (so $k[\mathbb{Z}/p\mathbb{Z}]$ indeed is not semisimple). Indeed, an irreducible representation of this group is a 1-dimensional space on which the generator acts by a pth root of unity. But every pth root of unity in k equals 1, as $x^p - 1 = (x-1)^p$ over k.

Problem 4.1.4. Let G be a group of order p^n. Show that every irreducible representation of G over a field k of characteristic p is trivial.

4.2. Characters

If V is a finite dimensional representation of a finite group G, then its character $\chi_V : G \to k$ is defined by the formula $\chi_V(g) = \mathrm{Tr}|_V(\rho(g))$. Obviously, $\chi_V(g)$ is simply the restriction of the character $\chi_V(a)$ of V as a representation of the algebra $A = k[G]$ to the basis $G \subset A$, so it carries exactly the same information. The character is a **central function**, or **class function**: $\chi_V(g)$ depends only on the conjugacy class of g; i.e., $\chi_V(hgh^{-1}) = \chi_V(g)$.

Denote by $F(G, k)$ the space of k-valued functions on G and by $F_c(G, k) \subset F(G, k)$ the subspace of class functions.

Theorem 4.2.1. *If the characteristic of k does not divide $|G|$, characters of irreducible representations of G form a basis in the space $F_c(G, k)$.*